Paper used:

Scott KY, Matthew R, Woolcock J, Silva M & Lemieux H (2019). Adjustments in the control of mitochondrial respiratory capacity to tolerate temperature fluctuations. *J Exp Biol* **222,** jeb207951.

Briefly, this paper investigates the thermal sensitivity of processes involved in oxidative phosphorylation (OXPHOS) in the planarian, *Dugesia tigrina*. High-resolution respirometry (HRR, which helps to look at different processes involved in respiration at the mitochondrial level) was used to measure activity of different stages of OXPHOS (activities of complex I, complex I+II, complex II, complex IV, and leak respiration), at five different temperatures (10, 15, 20, 25, and 30°C). Citrate synthase (marker of mitochondrial content) activity was also measured for these animals. Following this, the animals were acclimated to a cold (5°C) and an optimal temperature (22°C). Samples from the acclimation were subject to the same rounds of high-resolution respirometry and a citrate synthase assay at two temperatures, 10 and 20°C (maybe to be able to look at Q10 effects; 10°C is the lowest temperature that HRR can be conducted).
The study concluded that the NADH pathway (attributed to complex I activity) was reduced at low temperatures but there was an effect of acclimation where its activity was increased.

The results were analysed using multiple tests described in the paper (please refer to the paper for details!):

1. The authors checked for normality of the data to analyse it through either an ANOVA or a Kruskal-Wallis test. It intuitively seems like a bad idea to use two different models on different subsets of the same dataset. It also seems like a normality check on the data answers a different question than an ANOVA/Kruskal-Wallis test, so I don't agree that the results of one was related to the other.
2. They have used pairwise comparisons with Tukey test/Dunn comparisons. I assume all possible comparisons run are relevant to the study, because they haven't mentioned if any of these comparisons were planned before the experiment.

3. The different stages of OXPHOS are also continuous in nature, and dependent on the previous step, an ANOVA seems maybe reductive because it ignores that dependency.
4. During HRR, the person carrying out the respirometry usually makes judgement calls to collect slopes of data after a reagent is added. These are done in real-time within a period of maybe 5-10 minutes, and the selection of 'good' regions of the slope to collect data is set to the discretion of the experimenter, rather than pre-determined parameters to make that decision.

Given these inherent biases along with the not ideal data analysis presented in this paper, I would rephrase some the results presented.

1. "Cold temperature is associated with loss of NADH pathway capacity and increased limitation of OXPHOS-linked respiration by the phosphorylation system". I would have preferred to see a graph that directly compares each stage within each experimental temperature. I think this statement should read, "Cold temperatures seem to reduce NADH pathway capacity with some limitation of OXPHOS-linked respiration by the phosphorylation system". I think "loss" is a more dramatic word, with vague evidence to support it.
2. "Apparent complex IV excess capacity is small and is only detected at 10°C". I would have said, "Apparent complex IV excess capacity is small and was detected at 10°C". The use of the word "only" seems harsh, misleads the reader, and again, there is a lot of uncertainty associated with the data. I may have also presented the graphs on the log scale, to illustrate differences better.
3. There is also a lot of mention of "significance" based on p-values. I would have preferred if they had included some literature review on what effect sizes are meaningful in other HRR studies (maybe comparisons at more extreme temperatures, studies in different species, etc.).